

# Intervention Types

## EVENT Inject exogenous shock

Injects a market-wide disturbance.  
All agents react naturally.

- `supply_disruption` (magnitude=2.0)
- `demand_surge` (magnitude=1.5)
- `storage_crisis` (magnitude=2.5)
- `cost_reduction` (magnitude=0.5)
- `subsidy` (magnitude=1.0)

Tests: shock → X pathways

## TRAIT Override agent parameter

Fixes a parameter for one agent,  
a role, or a faction.

- All producers: `production_cost` = 0
- All consumers: `demand_value` = 200
- All producers: `storage_cost` = 100
- `producer_A`: `production_cost` = 0
- All consumers: `demand_per_period` = 0

Tests: parameter → output pathways

## ACTION Force specific agent order

Forces one agent to submit a  
specific order (price/quantity).

- `speculator_A`: `quantity` = 0
- `producer_A`: `limit_price` = 0
- `consumer_A`: `quantity` = 100
- `speculator_B`: `limit_price` = 500
- `producer_B`: `quantity` = 0

Tests: order flow → market outcomes

## Shared Mechanics

- Each intervention runs for 3 periods (configurable)
- A parallel baseline (no intervention) runs from the same starting state
- Results show per-variable deltas: intervention trajectory minus baseline trajectory
- Model sees: which variables changed, by how much, and in which direction